

Latin American Challenges and Recommendations for Poly Adenosine Diphosphate Ribose Polymerase Inhibitor Treatment in Metastatic Castration Resistant Prostate Cancer: An Expert Overview

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Abstract

In Latin America, prostate cancer is the third most common cancer overall and the most common in men, with the highest mortality rate of all cancers. In 2022, there were approximately 22,985 new prostate cancer cases and 61,056 deaths from prostate cancer in the region. Patients with metastatic disease that is resistant to cure by castration now have multiple therapeutic options, including poly-ADP ribose polymerase inhibitors. These treatment advances present new challenges, such as developing monitoring protocols for early detection of disease progression to castration resistance. The Americas Health Foundation organized a 3-day meeting with 8 regional oncologists and pathologists to create a paper on metastatic castration-resistant prostate cancer diagnosis and therapy, including the new poly-ADP ribose polymerase inhibitors. The panel examined metastatic castration-resistant prostate cancer in Latin America and recommended ways to improve patient care using published literature and their expertise. Gene mutations play an important role in prostate cancer development. Precision medicine innovations highlight the importance of genotyping DNA variants and tumor biomarkers for targeted treatment. Access to appropriate genetic testing is difficult, medications are available but expensive, and there is a lack of infrastructure and regulatory frameworks that prevent patients from benefiting from innovative therapies. The panel recommends developing a population database and biobank and creating tumor tissue collection, processing, and storage facilities. Multi-stakeholder collaboration is needed to integrate the information gathered, train staff, select target populations, improve patient accessibility, and reduce the cost burden of drugs, genetic counselors, and cancer geneticists in Latin America. Collaboration is essential

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among healthcare professionals, policymakers, patient advocacy groups, pharmaceutical companies, and international organizations to address these challenges and needs in Latin America.

Keywords

castration-resistant prostate cancer treatment, PARP inhibitor access in Latin America, PARP inhibitors for prostate cancer, prostate cancer in Latin America, prostate cancer treatment, targeted therapies for prostate cancer

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Introduction

In recent decades, LATAM has seen an increase in life expectancy.¹ This change leads to higher rates of cancer, especially prostate cancer (PCa).² Despite recent healthcare breakthroughs, cancer mortality rates in Latin America are double that of industrialized countries.³ PCa is the third most common cancer overall (following lung and breast) in the region and the most common among men, with the highest mortality rate of all cancers.

Several studies over the last decade have detailed the molecular landscape of advanced prostate cancer and the processes behind treatment resistance.⁴⁻⁶ Alterations in genes involved in the homologous recombination repair (HRR) pathway were found in one-quarter of patients with advanced prostate cancer,^{4,7} providing a unique opportunity to develop therapeutic techniques that take advantage of the tumor's diminished ability to repair DNA damage. Poly-ADP-ribose polymerase inhibitors (PARPi) have shown remarkable efficacy in treating patients with metastatic castration-resistant prostate cancer (mCRPC) with HRR abnormalities, especially in those with *BRCA1/2* mutations.^{5,7-11} However, budgetary restrictions, diverse demographics, a lack of qualified workers, and access barriers impede the adoption of PARPi in Latin America. This narrative review outlines the current landscape of poly-ADP ribose polymerase inhibitor (PARPi) treatment in LATAM and makes recommendations for improvement based on the literature, real-world experience, and expert opinions of the panel.

Methods

The Americas Health Foundation (AHF) assembled a panel of 8 specialists (pathologists and oncologists) who are experts in PCa from LATAM. The group met on June 27, 29, and 30, 2023, to develop recommendations to improve LATAM access to PCa testing and treatment. AHF used PubMed, MEDLINE, and EMBASE to identify LATAM pathologists and oncologists who had published articles on PCa diagnosis and treatment. All experts who attended the meeting are named authors of this manuscript.

AHF researched PCa in PubMed, MEDLINE, and EMBASE. The searches included "prostate cancer," "metastatic prostate cancer," and "prostate cancer treatment," in combination with "Latin America," from 01/01/2017 to 01/01/2023.

The identified articles were in English, Portuguese, and Spanish.

Based on the literature search, AHF developed specific questions to address barriers to metastatic castration-resistant prostate cancer (mCRPC) treatment in LATAM and assigned 1 to each panel member. Each panel member drafted a written response to their question based on the literature review and personal experience. The entire panel reviewed and edited each narrative during the 3-day conference through sufficient rounds of discussion until the panel reached a total agreement. An AHF staff member moderated the debate. When the panel disagreed, additional discussions were held until everyone agreed on the article's content. All authors reviewed and approved the final manuscript. The authors based their recommendations on evidence gathered and expert opinion; all authors approved the final paper.

Results

Prostate Cancer in Latin America

In 2022, the region had approximately 225,985 new cases and 61,056 deaths from PCa. Brazil has the highest number of reported cases of PCa, with 102,519 new cases and 61,056 deaths, followed by Mexico, with 26,565 cases. When considering the 5-year prevalence, Brazil has an incidence rate of 15.5/100,000, while Mexico's is 12.8/100,000. These statistics highlight the varying burden of PCa in LATAM and the Caribbean. Given the difficulties in collecting data in these nations, cases are likely under-reported. Argentina, Brazil, Colombia, Cuba, and other countries in the region face challenges in reducing PCa-related deaths.¹

Metastatic Castration-Resistant Prostate Cancer

Metastatic prostate cancer exhibits notable molecular and clinical heterogeneity. At the castration-resistant stage, it acquires an aggressive and incurable nature with a median overall survival (OS) of 2-3 years. Although estimating this population is difficult, a systematic review suggests that approximately 10%-20% of all prostate cancer patients will eventually develop mCRPC in 5 years.^{12,13}

Prognosis of mCRPC

The prognosis for patients with mCRPC has improved significantly in the past decade, thanks to innovative agents that improve OS. Patients now have different therapeutic options, including chemotherapeutic agents (docetaxel and cabazitaxel), second-generation androgen receptor inhibitors (ARPIs) (apalutimed, darolutamide, enzalutamide and abiraterone acetate), radiotherapeutic agents (radium 223 and lutetium-177), checkpoint inhibitors (pembrolizumab), and PARPi (olaparib, talazoparib, rucaparib, and niraparib).¹⁴⁻¹⁹

These advances present new challenges, such as developing monitoring protocols for early detection of disease progression and the transition from metastatic castration-sensitive PCa (mCSPC) to metastatic castration-resistant PCa (mCRPC). Furthermore, identifying biomarkers for personalized patient-drug matching is crucial for precision medicine. These efforts aim to reduce cancer treatment's adverse side effects and financial burden.

PARPi and mCRPC

PARPi were the first precision medicine approach to PCa. PARPi target the DNA damage repair (DDR) pathway, and their use has extended lives in men with mCRPC who harbor DDR mutations.^{16,18,19} PARPi are a targeted therapy aiming to

exploit differential DNA repair in malignant tumors. It induces objective tumor responses while improving progression-free and OS in men with mCRPC with HRR mutations. Table 1 highlights the pivotal trials for PARPi.²⁰⁻³⁴

In a retrospective study of 313 patients who died of PCa and 486 patients with low-risk localized PCa, *BRCA1/2*, and *ATM* mutations, the carrier rate was significantly higher in patients with lethal PCa (6.07% vs 1.44%).^{4,35,36} The PCa rate among *BRCA1* carriers was more than twice as high (8.6% vs 3.8%) as the general population.³⁷ Genetic aberrations of *BRCA1/2* and *ATM* genes are also significantly higher (19.3%) in patients with metastatic disease than those with advanced localized PCa (5%).^{4,38} Approximately 20%-25% of patients with metastatic PCa have gene mutations directly or indirectly related to DNA repair by the HRR mechanism—50% are germline.^{4,35,39}

In PCa, germline *BRCA1/2* mutations are associated with a higher probability of a Gleason score ≥ 8 , greater regional lymph node involvement, and worse prognosis in localized disease compared to patients with wild-type *BRCA1/2*. Pathogenic variants of *BRCA1/2* are associated with early disease onset, aggressive tumors, higher recurrence, and poor prognosis; therefore, timely identification by genetic testing becomes compelling.^{4,35,36} A prospective cohort study of male *BRCA1/2* variant carriers confirmed the association of *BRCA2* with aggressive PCa.⁴⁰ An analysis of the results of 2019 PCa patients

Table 1. Published Phase III Trials With PARPi Monotherapy and in Combination With ARPi.

	PROFOUND	TRITON-3	MAGNITUDE	PROPEL	TALAPRO-2
Control Arm	Abiraterone or Enzalutamide	Docetaxel or Abiraterone or Enzalutamide	Abiraterone acetate + Prednisone + Placebo	Abiraterone + Placebo	Enzalutamide + Placebo
Experimental Arm	Olaparib	Rucaparib	Niraparib + Abiraterone acetate + Prednisone	Olaparib	Talazoparib + Enzalutamide
Population	Cohort A (<i>BRCA1/2</i> and <i>ATM</i>) Cohort B (other HRRm)	<i>BRCA1/2</i> and <i>ATM</i>	Cohort HRRm Cohort HRRnm vs Halted due to futility	HRRm 28% of population	HRRm 21% of population
rPFS All Corners	-----	-----	-----	24.8 vs 16.6 months HR 0.66 $P < 0.001$	NR vs 21.1 months HR 0.63 $P < 0.001$
rPFS <i>BRCA1/2</i> Subgroup	-----	10.2 vs 6.4 months HRRm 0.61 $P = 0.0003$	16.6 vs 10.9 months HRRm 0.53 $P = 0.0014$	HRRnm 8.4 months HRRm 0.23 (95% CI 0.12-0.43)	Not reported
rPFS HRRm Subgroup	7.39 vs 3.55 months HRRm 0.34 $P < 0.0001$ (Cohort A)	8.1 vs 6.8 months (ATM) HRRm 0.95 $P = 0.84$	16.5 vs 13.7 months HRRm 0.73 $P = 0.0217$	HRRnm 13.9 months HRRm 0.50 (95% CI 0.34-0.73)	27.9 vs 16.4 months HRRm 0.46 (95% CI 0.3-0.7)
rPFS Non-HRRm/Unknown	-----	-----	-----	24.1 vs 19 months HRRm 0.76 (95% CI 0.60-0.97)	HRRnm 22.5 months HRRm 0.70 (95% CI 0.58-0.89)
ORR	33.3% vs 2.3% $P < 0.0001$	-----	60% vs 28% $P < 0.001$	58% vs 48% $P = 0.041$	62% vs 44% $P = 0.005$

rPFS, radiographic progression-free survival; HRRm, homologous recombination repair mutation; HRRnm, non-homologous recombination repair mutation.

(18 *BRCA1*-variant carriers, 61 *BRCA2*-variant carriers, and 1940 non-carriers) showed that PCa with germline *BRCA1/2* mutations were more frequently associated with International Society of Urological Pathology (ISUP) score greater than 4, T3/T4 stage, nodal involvement and metastases at diagnosis.⁴¹ *BRCA*-susceptibility gene mutations were also reported to have worse outcomes than non-carriers after local therapy.⁴²

In patients with metastatic PCa, the efficacy of androgen deprivation therapy is lower in patients with germline *BRCA1/2* mutations.^{41,42} Mutations in HRR genes have shown sensitivity to PARPi in various tumors (breast, ovarian, prostate, and pancreas). These mutations can affect a patient's response to treatment. (Table 1) Several phase II studies with PARPi as monotherapy have shown high objective response rates (ORR) in patients with *BRCA1/2* mutations and mCRPC.^{7,10,11,36} (Figure 1).

Suggested treatment algorithm for mCRPC. Figure 2 describes the suggested treatment algorithm based on the information gleaned from experience and published clinical trials. Patients with advanced PCa should undergo a biopsy to obtain histologic confirmation at the time of diagnosis and later if needed. Conventional imaging tests (bone scintigraphy, CT,

MRI) should be ordered for mCSPC at any sign of progression (including clinical symptoms or elevated PSA). For patients with stable PSA and no signs of clinical progression, imaging tests could be ordered once a year. Today, there is no recommendation for monitoring or changing treatment based on PSMA PET. The panel recommends personalized treatment with a PARPi in patients with a *BRCA* pathogenic variant. When available, every patient with mPC should undergo germline testing, although the results should only drive therapy for patients with mCRPC.

ARPi vs ARPi + PARPi

Phase III trials have raised discussions about the clinical applicability and cost-effectiveness of ARPi + PARPi combinations, particularly in patients with wild-type HRR genes, compared to those with *BRCA1/2* mutations (Table 2). The feasibility of applying these results to current mCSPC patients who previously received ARPi in the hormone-sensitive setting remains a question of interest.

The decision to combine an ARPi with a PARPi for patients without HRR mutations may be guided by clinical parameters

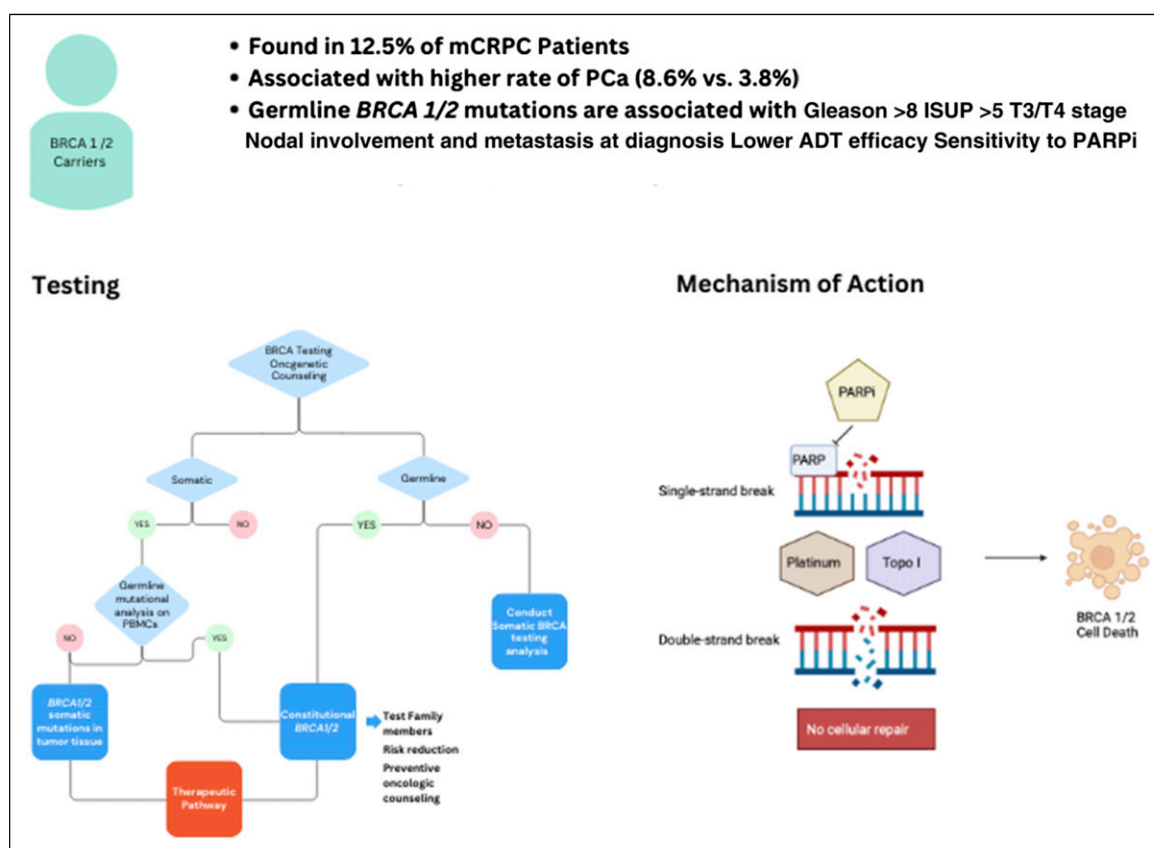


Figure 1. *BRCA 1/2* mutation in prostate cancer. Traits, diagnosis, and mechanisms of action of PARPi. mCRPC, metastatic castration-resistant prostate cancer; PCa, prostate cancer; ISUP, International Society of Urological Pathology; ADT, androgen deprivation therapy; PARPi, poly-ADP ribose polymerase inhibitors.

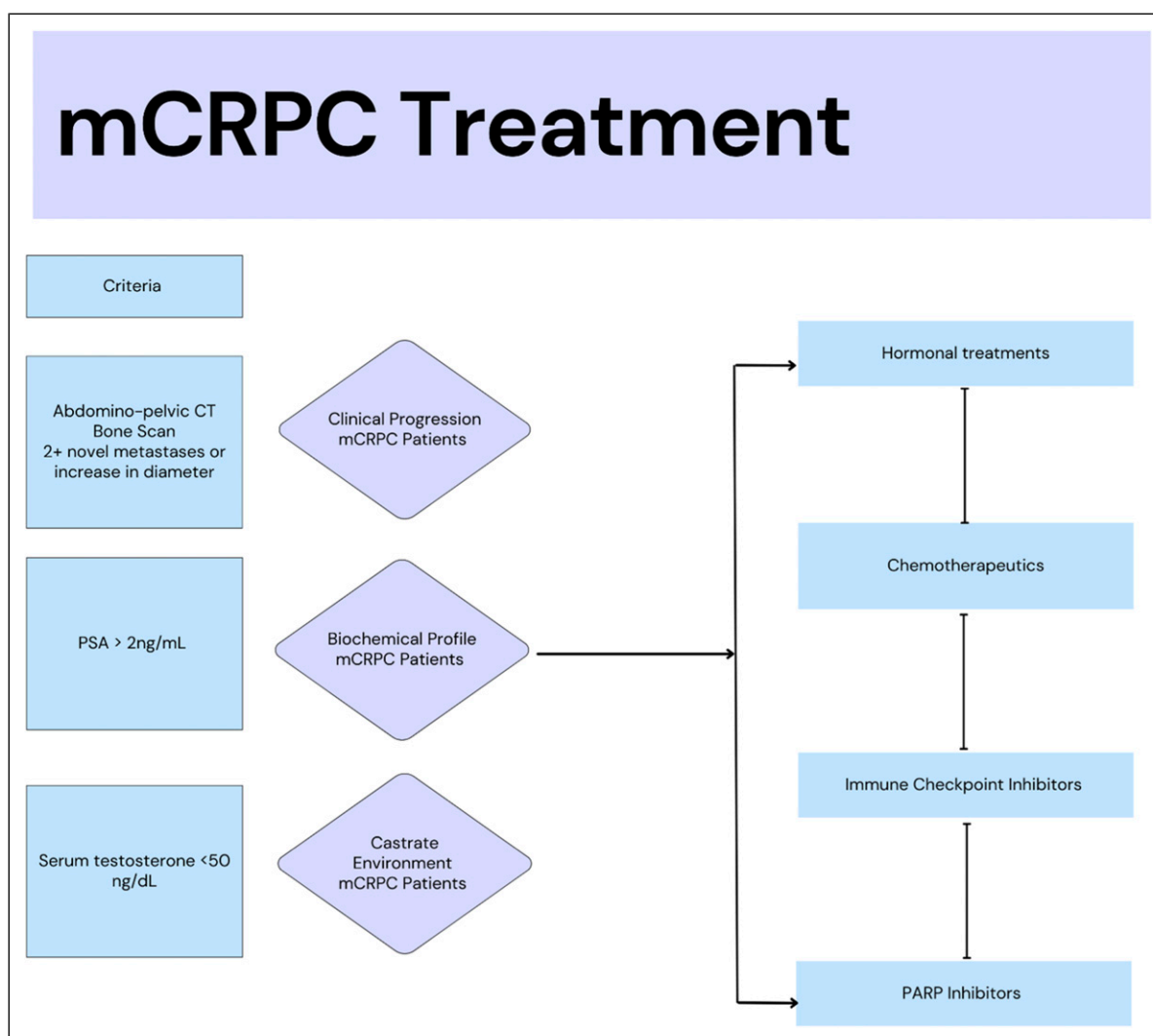


Figure 2. Treatment algorithm for metastatic castration-resistant prostate cancer.

such as (but not limited to) age, extent of visceral involvement, and aggressiveness and rapid progression of the disease. HRR gene status predicts the response to PARPi and provides prognostic information, with worse outcomes associated with HRR mutations. Additionally, HRR gene testing can serve as a basis for genetic counseling for germline mutations.

PARPi safety. These drugs are safe with manageable side effects, mainly hematologic (anemia) and digestive (nausea, vomiting, diarrhea). Talazoparib has the highest incidence of hematologic toxicity (approximately 40% experiencing grade 3/4 anemia) among PARPi, while olaparib and niraparib have slightly higher rates of gastrointestinal adverse events. Quality of life remains unaffected in Phase III studies, confirming the favorable profile of these drugs. Rare adverse events include myelodysplastic syndrome and acute myeloid leukemia.

PARPi availability in Latin America. In Argentina, olaparib is the only approved PARPi for mCRPC patients with HRR gene

mutations who progressed on second-generation ARPi (abiraterone and/or enzalutamide). Other PARPi (rucaparib) or combination therapies (talazoparib + enzalutamide; niraparib + abiraterone; olaparib + abiraterone) are yet to receive regulatory approval. In Chile, olaparib is approved for patients with *BRCA1/2* mutations, but mutation testing and treatment are not supported in the public healthcare system and are partially covered in private healthcare systems (Table 3).

Discussion

Limitations and Recommendations for Improving Appropriate Access to PARPi for MCRPC in Latin America

LATAM's health systems face diverse challenges in diagnosing and intervening early in mCRPC. Some challenges include inequitable access to testing in the public healthcare sector, biomarker screening and its meaning in the ethnically

Table 2. Access and Cost of HRR Testing and PARPi for mCRPC.

	PARPi Available	Coverage of Treatment and HRR Testing		Costs (USD/month)		Barriers to Accessing Proper Diagnosis
		Public	Private	Treatment	HRR Tests	
Argentina	Olaparib	NO	YES	3000-4000	400	Regional challenges
Brazil	Olaparib	NO	NO ^a	5000-6000	800	Regional challenges, poor access in rural areas
Chile	Olaparib	NO	PARTIAL	6000	600	CT and bone scans are difficult to access, high treatment costs, long wait times
Colombia	Olaparib	YES ^b	YES	4000-5000	400	Regional challenges in access, correct drug delivery
Mexico	Olaparib	YES ^c	YES	3000-4000	700	Diagnostics are offered; however, rural access is unreliable

^aInsurance companies will provide testing where needed to determine the appropriate treatment.

^bOlaparib is approved as monotherapy after progressing on 1 ARPi.

^cWith restrictions on availability.

Table 3. Effect of *BRCA1/2* Mutations on PARPi Efficacy in Treating mCRPC (Phase III Clinical Trials).

Clinical Trial	Line (N)	PARPi	BRCAm	HRRm	Non-HRRm
PROfound*	2 nd (367)	Olaparib	0.22	0.49	N/A
TRITON-3**	2 nd (405)	Rucaparib	0.50	0.61 (BRCA + ATM)	N/A
Magnitude**	1 st (670)	Niraparib	0.55	0.76	Stopped for futility
TALAPRO-2**	1 st (1095)	Talazoparib	—	0.46	0.69
PROpel	1 st (796)	Olaparib	0.23	0.50	0.76

BRCAm, mutation in *BRCA1* or *BRCA2*; HRRm, homologous recombination repair mutation; Non-HRRm, non-homologous recombination repair mutation.

*sNDA olaparib submission; **ASCO GU 2023.

diverse region, a lack of oncologists and skilled researchers, high-cost tests and drugs, and fragmented regulatory and healthcare frameworks. Improving access to healthcare services, including the use of telehealth and artificial intelligence (AI), is vital, particularly in underserved rural areas (Figure 3).

Imaging In Metastatic Prostate Cancer

Intensive monitoring protocols used in pivotal trials cannot be adopted in everyday clinical practice. There is still no agreement on the frequency or methods of monitoring patients with castration-sensitive PCa (CSPC) and castration-resistant PCa (CRPC) during the evolution of their disease among the guidelines issued by the European Association of Urology, the American Urology Association, the European Society of Medical Oncology, and the National Comprehensive Cancer Network.^{17,20-22}

Advanced imaging techniques such as choline or prostate-specific membrane antigen (PSMA) PET and whole-body MRI have introduced complexity in evaluating PCa. These methods demonstrate greater sensitivity than traditional bone scintigraphy and CT scans, particularly in assessing skeletal involvement and detecting bone and soft tissue metastases. However, there is no consensus on evaluating and categorizing test results in advanced PCa.

In LATAM, there is a significant disparity between the availability of imaging methods in public and private healthcare

systems. PSMA PET is often unavailable in the public system, and conventional imaging, when available, is often inadequate. Although PSMA PET has been investigated to monitor therapeutic responses in patients with mCSPC, we cannot yet recommend it for disease management because of limited evidence. Current clinical trials primarily use conventional imaging to evaluate disease progression and tumor biology.²³

Recommendations

Government. Contribute to reducing the time it takes to make a diagnosis by improving the availability and accessibility of high-quality conventional imaging methods for patients with mCRPC.

Medical Societies. Provide guidelines and standards for diagnostic imaging and advocate for necessary resources and training.

Academic and Research Institutions. Develop and disseminate advanced imaging protocols and conduct research that influences policymakers and healthcare providers.

Biomarkers

PCa exhibits molecular diversity and requires precision treatment strategies. Molecular sequencing and liquid biopsies can potentially identify predictive biomarkers for guiding clinical decisions.

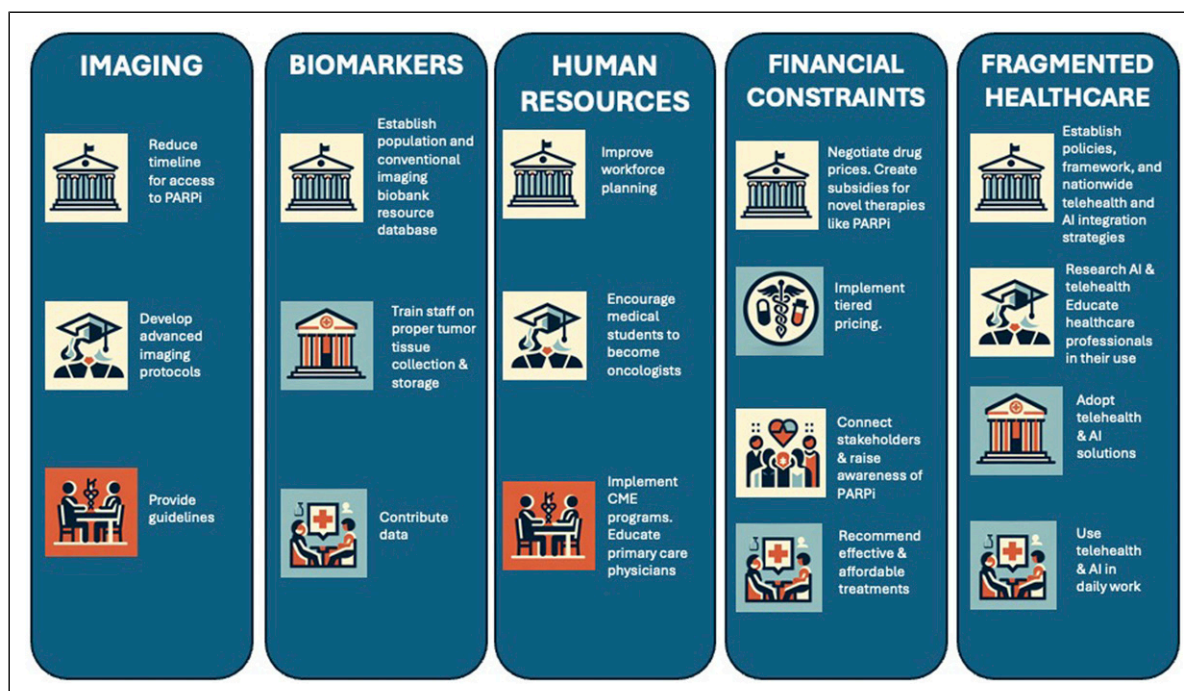


Figure 3. Recommendations overview with responsible stakeholders. Icons.

Recent studies highlight the importance of identifying HRR mutations,⁴³ both somatic and inherited. A viable biomaterial is essential, with tumor tissue preferred for detecting somatic variants and blood preferred for germline variants.^{44,45} However, inadequate tissue processing and/or storage often hamper obtaining sufficient, high-quality nucleic acid from samples.²⁴⁻²⁶

Characterizing the frequency of cancer-predisposing gene mutations is the first step in developing meaningful biomarkers and a precision approach to mCRPC. Latin American countries exhibit dramatic diversity due to the genetic combination of various ethnic groups and lifestyles.^{46,47} This genetic mixing results in disparities in treatment responses among different groups (pharmacoethnicity).⁴⁸

Coverage of PCa genetic testing in LATAM is limited. In Brazil, molecular testing of HRR genes is only available for rare diseases and is not yet accessible for mCRPC. In Colombia, testing is covered by the public healthcare system and, sometimes, paid for by the pharmaceutical industry. In some countries, the pharmaceutical industry covers HRR gene testing in research contexts and for PARPi clinical trials.

Recommendations

Government. Establish a comprehensive and reliable population and conventional imaging biobank resource database. The governments should share the information with other countries in the region and provide the necessary infrastructure for housing and maintaining the biobanks.

Hospitals. Train their staff on proper tumor tissue collection, processing, and storage.

Healthcare Professionals. Contribute accurate and high-quality patient data.

All stakeholders should integrate the information to develop a cost-effective genetic testing program to select the appropriate target population of people with mCRPC who will benefit from PARPi therapy.

Human Resources

A lack of trained personnel impedes timely diagnosis and treatment of mCRPC.⁴⁹ LATAM has a low number of trained oncologists compared to the number of new cancer cases.⁵⁰ (Table 4). Moreover, residents need training in precision medicine, and continuing education is necessary for oncologists to take full advantage of PARPi and other targeted treatments for appropriate patients.

Recommendations

Government. Improve workforce planning to support more opportunities for oncology training.

Universities and Colleges. Recognize the need for oncologists and communicate this need to incoming medical students.

Medical Societies. Implement continuing medical education programs to ensure that people with mCRPC are appropriately recognized, diagnosed, and treated. Educate primary care physicians about the importance of identifying and referring patients at risk of PCa to specialists.

Table 4. Cancer Rates and Number of Oncologists in Seven Latin American Countries.

Country	Annual Cancer Incidence	Annual Cancer Mortality	Mortality-To-Incidence Ratio	No. of Clinical Oncologists	Ratio of New Cancer Cases/Clinical Oncologist
Argentina	115,000	66,000	0.57	400	287
Brazil	438,000	225,000	0.51	2577	170
Chile	40,000	25,000	0.63	60	667
Mexico	148,000	79,000	0.53	352	420
Panama	5400	2900	0.54	10	540
Peru	43,000	26,000	0.60	130	331
Uruguay	13,000	9000	0.69	120	108

Financial Constraints

LATAM has experienced a significant increase in cancer drug costs because of the shift toward targeted therapies. Some countries, including Argentina and Mexico, have explored reinsurance options for “low-incidence and high-cost” diseases. However, income inequality and poverty present substantial challenges. In Argentina, healthcare access varies widely, impacting patients’ ability to receive optimal treatment. Academic reference centers have adopted targeted treatments like olaparib for PCa, benefiting from multidisciplinary teams and continuing education.⁵¹ Despite efforts to increase financial protection, projected demands for legalization from beneficiaries and disputes from those responsible for providing the claimed benefits make it challenging.

Recommendations

Government. Negotiate drug prices, regulate the pharmaceutical industry to prevent excessive pricing, and create reimbursement and subsidies to help offset the higher costs of targeted therapies like PARPi.

Pharmaceutical Companies. Implement tiered pricing, offer patient assistance programs, and collaborate with the government to reduce costs in high-poverty regions.

Patient Advocacy Groups. Serve as intermediaries among patients, healthcare providers, and policymakers. Raise awareness about financial toxicity in cancer survivors. Advocate for more affordable diagnostics and treatment.

Healthcare Providers. Recommend treatments that are both effective and affordable.

Fragmented Healthcare Systems

Healthcare across LATAM is disjointed and often inequitable. Public healthcare systems, which support the majority of people living in LATAM, are often underfunded, leading to outdated equipment, drug shortages, and long wait times. Urban areas typically have more healthcare professionals and better facilities than rural areas. While addressing the whole

problem is beyond the scope of this paper, incorporating 2 newer technologies, telehealth and AI, can help alleviate some of the inequity.

Telehealth can connect patients in underserved and rural areas with specialists, enhance collaboration among healthcare providers across the region, and provide educational resources and support for patients with mCRPC. It can also help with patient care continuity, facilitating dialogue regarding side effects or complications. By reducing travel needs, telehealth can reduce the overall cost of their healthcare and make advanced therapies more accessible.

AI has the potential to tackle multiple challenges and enhance access to targeted therapies such as PARPi. By utilizing AI algorithms, patient data can be analyzed to customize treatments according to individual genetic profiles. This has the potential to enhance the effectiveness of PARPi and other therapies. Additionally, AI-driven chatbots and virtual assistants can assist telehealth services by offering initial diagnoses, addressing patient inquiries, and scheduling appointments. This expands the availability of healthcare services.

Furthermore, the combination of AI and telehealth can provide personalized training programs for healthcare providers, ensuring that they stay abreast of the latest developments in treatment protocols and regulatory requirements. Stakeholders must also consider the unique challenges faced by Latin American countries, such as varied economic conditions, differing healthcare infrastructure, and disparities in access to technology. Tailoring AI implementations to address these regional nuances is vital for success.

Recommendations

Government. Establish policies, framework, and nationwide telehealth and AI integration strategies.

Universities and Colleges. Research AI and telehealth use in healthcare to monitor effectiveness and innovate new technologies. Educate healthcare professionals in AI and telehealth.

Hospitals. Adopt telehealth and AI solutions for patient consultation, diagnosis, and follow-up care.

Healthcare Providers. Provide medical advice, consultations, and treatment plans via telehealth platforms. Incorporate AI into everyday tasks such as diagnosis, treatment, and patient management.

Conclusions

Addressing limited access to innovative medications like PARPi requires collaborative solutions among stakeholders, including policymakers, government agencies, healthcare providers, hospitals, patient advocacy groups, medical societies, private institutions, and medical schools and universities. Non-governmental organizations can help bridge gaps in resources and public outreach. Collaborations with international groups like the WHO Pan American Health Organization can assist in training, expertise, and other resources. This collaboration can improve the region's management and treatment outcomes of mCRPC. Incorporating education programs for the population, implementing public policies for access to health services, biomarkers, and telemedicine, and using AI in imaging for high-risk patients are necessary steps forward.

Appendix

Table of Abbreviations

ADT	Androgen deprivation therapy
AHF	Americas Health Foundation
CRPC	Castration-resistant PCa
CSPC	Castration-sensitive PCa
DDR	DNA damage repair
U.S. FDA	Food and Drug Administration
HR	Hazard ratio
HRR	Homologous recombination repair
HRRm	Homologous recombination repair mutation
HRRmn	Non-homologous recombination repair mutation
ISUP	International Society of Urological Pathology
LATAM	Latin America
ORR	Objective response rate
OS	Overall survival
PFS	Progression-free survival
PSA	Prostate-specific antigen
rPFS	Radiographic progression-free survival

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References

1. International Agency for Research on Cancer, World health center. <https://gco.iarc.fr/en>
2. Bray F, Piñeros M. Cancer patterns, trends and projections in Latin America and the Caribbean: a global context. *Salud Publica Mex.* 2016;58:104-117.
3. Strasser-Weippl K, Chavarri-Guerra Y, Villarreal-Garza C, et al. Progress and remaining challenges for cancer control in Latin America and the Caribbean. *Lancet Oncol.* 2015;16:1405-1438.
4. Robinson D, Van Allen EM, Wu Y-M, et al. Integrative clinical genomics of advanced prostate cancer. *Cell.* 2015;162:454.
5. Mateo J, Seed G, Bertan C, et al. Genomics of lethal prostate cancer at diagnosis and castration resistance. *J Clin Invest.* 2020;130:1743-1751.
6. Armenia J, Wankowicz SAM, Liu D, et al. The long tail of oncogenic drivers in prostate cancer. *Nat Genet.* 2018;50:645-651.
7. de Bono JS, Mehra N, Scagliotti GV, et al. Talazoparib monotherapy in metastatic castration-resistant prostate cancer with DNA repair alterations (TALAPRO-1): an open-label, phase 2 trial. *Lancet Oncol.* 2021;22:1250-1264.
8. Mirza MR, Monk BJ, Herrstedt J, et al. Niraparib maintenance therapy in platinum-sensitive, recurrent ovarian cancer. *N Engl J Med.* 2016;375:2154-2164.
9. González-Martín A, Pothuri B, Vergote I, et al. Niraparib in patients with newly diagnosed advanced ovarian cancer. *N Engl J Med.* 2019;381:2391-2402.
10. Mateo J, Porta N, Bianchini D, et al. Olaparib in patients with metastatic castration-resistant prostate cancer with DNA repair gene aberrations (TOPARP-B): a multicentre, open-label, randomised, phase 2 trial. *Lancet Oncol.* 2020;21:162-174.
11. Abida W, Patnaik A, Campbell D, et al. Rucaparib in men with metastatic castration-resistant prostate cancer harboring a BRCA1 or BRCA2 gene alteration. *J Clin Oncol.* 2020;38:3763-3772.
12. Scher HI, Solo K, Valant J, et al. Prevalence of prostate cancer clinical states and mortality in the United States: estimates using a dynamic progression model. *PLoS One.* 2015;10:e0139440.
13. Comford P, Bellmunt J, Bolla M, et al. EAU-ESTRO-SIOG guidelines on prostate cancer. Part II: treatment of relapsing, metastatic, and castration-resistant prostate cancer. *Eur Urol.* 2017;71:630-642.
14. Turco F, Gillesen S, Cathomas R, et al. Treatment landscape for patients with castration-resistant prostate cancer: patient selection and unmet clinical needs. *Res Rep Urol.* 2022;14:339-350.
15. Sartor O, De Bono J, Chi KN, et al. Lutetium-177-PSMA-617 for metastatic castration-resistant prostate cancer. *N Engl J Med.* 2021;385:1091-1103.
16. de Bono J, Mateo J, Fizazi K, et al. Olaparib for metastatic castration-resistant prostate cancer. *N Engl J Med.* 2020;382:2091-2102.

17. Chi KN, Rathkopf D, Smith MR, et al. Niraparib and abiraterone acetate for metastatic castration-resistant prostate cancer. *J Clin Oncol.* 2023;41:3339-3351.
18. Agarwal N, Azad AA, Carles J, et al. Talazoparib plus enzalutamide in men with first-line metastatic castration-resistant prostate cancer (TALAPRO-2): a randomised, placebo-controlled, phase 3 trial. *Lancet.* 2023.
19. Abida W, Campbell D, Patnaik A, et al. Rucaparib for the treatment of metastatic castration-resistant prostate cancer associated with a DNA damage repair gene alteration: final results from the phase 2 TRITON2 study. *Eur Urol.* 2023;S0302:02828.
20. Parker C, Castro E, Fizazi K, et al. Clinical practice guidelines – prostate cancer. *Ann Oncol.* 2020.
21. Mottet N, Van den Bergh RCN, Briers E, et al. Eau-canm-estro-esuriog guidelines on prostate cancer. *Eur. Assoc. Urol.* 2020;1:11-143.
22. Lowrance WT, Breaux RH, Chou R, et al. Advanced prostate cancer: AUA/ASTRO/SUO guideline PART I. *J Urol.* 2021; 205:14-21.
23. Atiq MO, Gandhi SU, Karzai F, et al. *PSMA PET Findings in Patients with Undetectable PSA More than 3 Years after Docetaxel for Metastatic Castration-Sensitive Prostate Cancer (mCSPC).* American Society of Clinical Oncology; 2022.
24. Seifert BA, O'Daniel JM, Amin K, et al. Germline analysis from tumor–germline sequencing dyads to identify clinically actionable secondary findings. *Clin Cancer Res.* 2016;22:4087-4094.
25. Scher HI, Morris MJ, Stadler WM, et al. Trial design and objectives for castration-resistant prostate cancer: updated recommendations from the Prostate Cancer Clinical Trials Working Group 3. *J Clin Oncol.* 2016;34:1402-1418.
26. Meric-Bernstam F, Brusco L, Daniels M, et al. Incidental germline variants in 1000 advanced cancers on a prospective somatic genomic profiling protocol. *Ann Oncol.* 2016;27:795-800.
27. Hubbard GK, Mutton LN, Khalili M, et al. Combined MYC activation and Pten loss are sufficient to create genomic instability and lethal metastatic prostate cancer. *Cancer Res.* 2016;76:283-292.
28. Gurel B, Iwata T, M Koh C, et al. Nuclear MYC protein overexpression is an early alteration in human prostate carcinogenesis. *Mod Pathol.* 2008;21:1156-1167.
29. Zou M, Toivanen R, Mitrofanova A, et al. Transdifferentiation as a mechanism of treatment resistance in a mouse model of castration-resistant prostate cancer. *Cancer Discov.* 2017;7:736-749.
30. Park JW, Lee JK, Sheu KM, et al. Reprogramming normal human epithelial tissues to a common, lethal neuroendocrine cancer lineage. *Science.* 2018;362:91-95.
31. Mu P, Zhang Z, Benelli M, et al. SOX2 promotes lineage plasticity and antiandrogen resistance in TP53-and RB1-deficient prostate cancer. *Science.* 2017;355:84-88.
32. Ku SY, Rosario S, Wang Y, et al. Rb1 and Trp53 cooperate to suppress prostate cancer lineage plasticity, metastasis, and antiandrogen resistance. *Science.* 2017;355:78-83.
33. Stopsack KH, Nandakumar S, Wibmer AG, et al. Oncogenic genomic alterations, clinical phenotypes, and outcomes in metastatic castration-sensitive prostate cancer. *Clin Cancer Res.* 2020;26:3230-3238.
34. Robinson D, Van Allen EM, Wu Y-M, et al. Integrative clinical genomics of advanced prostate cancer. *Cell.* 2015;161:1215-1228.
35. Pritchard CC, Mateo J, Walsh MF, et al. Inherited DNA-repair gene mutations in men with metastatic prostate cancer. *N Engl J Med.* 2016;375:443-453.
36. Mateo J, Carreira S, Sandhu S, et al. DNA-repair defects and olaparib in metastatic prostate cancer. *N Engl J Med.* 2015;373:1697-1708.
37. Na R, Zheng SL, Han M, et al. Germline mutations in ATM and BRCA1/2 distinguish risk for lethal and indolent prostate cancer and are associated with early age at death. *Eur Urol.* 2017;71:740-747.
38. Kote-Jarai Z, Leongamornlert D, Saunders E, et al. BRCA2 is a moderate penetrance gene contributing to young-onset prostate cancer: implications for genetic testing in prostate cancer patients. *Br J Cancer.* 2011;105:1230-1234.
39. Abida W, Armenia J, Gopalan A, et al. Prospective genomic profiling of prostate cancer across disease states reveals germline and somatic alterations that may affect clinical decision making. *JCO precision oncology.* 2017. PO-17, 2017.
40. Nyberg T, Frost D, Barrowdale D, et al. Prostate cancer risks for male BRCA1 and BRCA2 mutation carriers: a prospective cohort study. *Eur Urol.* 2020;77:24-35.
41. Castro E, Goh C, Olmos D, et al. Germline BRCA mutations are associated with higher risk of nodal involvement, distant metastasis, and poor survival outcomes in prostate cancer. *J Clin Oncol.* 2013;31:1748-1757.
42. Castro E, Goh C, Leongamornlert D, et al. Effect of BRCA mutations on metastatic relapse and cause-specific survival after radical treatment for localised prostate cancer. *Eur Urol.* 2015;68:186-193.
43. Fizazi K, Piulats JM, Reaume MN, et al. Rucaparib or physician's choice in metastatic prostate cancer. *N Engl J Med.* 2023; 388:719-732.
44. Polkinghorn WR, Parker JS, Lee MX, et al. Androgen receptor signaling regulates DNA repair in prostate cancers. *Cancer Discov.* 2013;3:1245-1253.
45. Li L, Karanika S, Yang G, et al. Androgen receptor inhibitor-induced "BRCAness" and PARP inhibition are synthetically lethal for castration-resistant prostate cancer. *Sci Signal.* 2017;10:eaam7479.
46. Forman D, Sierra MS. Cancer in central and south America: introduction. *Cancer Epidemiol.* 2016;44(Suppl 1):S3-s10.
47. Avena S, Via M, Ziv E, et al. Heterogeneity in genetic admixture across different regions of Argentina. *PLoS One.* 2012;7:e34695.
48. O'Donnell P, Me D. Pharmacoethnicity: ethnic differences in susceptibility to the effects of chemotherapy. *Clin Cancer Res.* 2009;15:4806-4814.
49. Global regional. National life expectancy, all-cause mortality, and cause-specific mortality for 249 causes of death, 1980-2015: a systematic analysis for the Global Burden of Disease Study 2015. *Lancet.* 2016;388:1459-1544.
50. Mathew A. Global survey of clinical oncology workforce. *J Glob Oncol.* 2018;4:1-12.
51. Williams SB, Kamat AM. The multidisciplinary approach to prostate cancer management: from diagnosis and beyond. *Everyday Urology - Oncology Insights.* 2017;2.